

Deliverable 5: Exception Flow Specification Document

Course: Software Engineering Lab 1 (PES University, Dept. of CSE)

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Problem Statement #05: Academic Elective Bidding & Allocation System

1. Exception Flow Summary Matrix

Exception ID	Target UC	Exception Title	Error Category	Severity	Detection & Action
EF-01	UC-01	Credit Sum Mismatch (!= 100)	Data Validation	High	Form sum != 100. System halts commit, displays dynamic point counter, highlights fields.
EF-02	UC-01 / UC-02	Unmet Prerequisite Check Failure	Domain Rule	High	Prerequisite missing. System rejects line item bid, flags transcript gap, prompts swap/waiver.
EF-03	UC-01 / UC-03	Unresolvable Slot Collision	Hard Constraint	High	Preference overlaps core slot. System rejects matrix & prompts non-conflicting choice.
EF-04	UC-01	Deadline Expiry Lockout	Temporal / State	Critical	Timestamp > deadline. System aborts transaction, enforces READ_ONLY, preserves prior state.
EF-05	UC-05	Solver Convergence / Timeout	Performance	Critical	Execution > 30s limit. Watchdog kills process, rolls back DB checkpoint, alerts Registrar.
EF-06	UC-01 / UC-05	DB Concurrency Deadlock	Infrastructure	High	Catch DB lock. Execute idempotent exponential retries; enqueue bid request if load spikes.
EF-07	UC-07	Unauthorized Override Attempt	Security / RBAC	Critical	Missing ROLE_REGISTRAR. System returns 403 Forbidden, logs immutable audit entry, flags origin IP.

2. System Resilience & Rollback Matrix

Exception	Recovery Strategy	Data Integrity Action	User / Admin Action Required
EF-01	Client/Server Form Hold	Uncommitted; draft preserved locally.	Student adjusts point totals to 100.
EF-02	Line-Item Rejection	Uncommitted for ineligible course.	Student picks eligible course / requests waiver.
EF-03	Conflict Interception	Uncommitted; prior state retained.	Student selects alternate non-clashing slot.
EF-04	Atomic Window Seal	Transaction rolled back; read-only lock.	View final locked pre-deadline submission.
EF-05	Watchdog Execution Kill	Full DB Rollback to pre-run checkpoint.	Registrar relaxes soft constraints & re-runs.
EF-06	Idempotent Retry & Queue	Eventual Consistency via Message Queue.	None (automatic retry) or short wait.
EF-07	Session Terminate & Ban	Audit Log Locked (Write-Once Access).	System Security Admin notified.